Project: Worked examples
Title: Insulation thickness (ISO 12241 formula route)
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1. Scope and method

The insulation thickness on a pipe (cylinder) or a flat surface that meets a heat-flow limit, a limit on the surface temperature of a hot surface and an anti-condensation requirement for a cold one, each solved separately and the governing one reported; and for a chosen thickness the heat flow, the surface temperature and PASS/FAIL per criterion. Steady state, one layer, no thickness table of any standard. Pipe resistance per metre: R = ln(D_e/D_i)/(2π·λ) + 1/(h_se·π·D_e), D_e = D_i + 2t; with the inner film and pipe wall R_oth = 1/(h_i·π·D_p) + ln(D_i/D_p)/(2π·λ_w) added in series Flat resistance per square metre: R = t/λ + 1/h_se (+ 1/h_i + t_w/λ_w) Heat flow: q = (θ_i − θ_a)/R, positive from the process to the ambient Surface temperature: θ_se = θ_a + q/(h_se·π·D_e) (pipe), θ_a + q/h_se (flat) Surface coefficient: h_se as given, or h_se = h_c + h_r with h_r = ε·σ·(T_s² + T_a²)(T_s + T_a); h_c from free convection to air (laminar or turbulent, the larger), forced convection (cross-flow cylinder or flat plate) and their cube-law blend; solved with θ_se, which it depends on Heat-loss limit: the thickness beyond which |q| ≤ q_max; surface-temperature limit (hot): θ_se ≤ θ_se,max; anti-condensation (cold): θ_se ≥ θ_dew + margin, θ_dew entered (for example from B2) or from the Magnus relation at θ_a and RH Solve: each criterion's thickness by scanning t from 0 to 1 m for the last broken point and bisecting the bracket after it; governing t = max over the selected criteria Critical diameter: D_crit = 2λ/h_se; below it, insulation raises the heat flow until D_e passes D_crit

Ref: ISO 12241 formula route (clause numbers to be checked) — Thermal insulation for building equipment and industrial installations: calculation rules
Ref: Incropera, Fundamentals of Heat and Mass Transfer 7th ed., §7.2, §7.4, §9.6, §9.9 (to be checked) — External forced, free and mixed convection correlations
Ref: ISO 13788 cl. 6 (clause number to be checked against the owner's copy) — Magnus saturation vapour pressure (shared with B7)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
geometry	Surface	pipe (cylinder)		assumption: A round pipe.	any
D_i	Pipe outer diameter, the inside of the insulation (pipe)	114	mm	drawing: M-PIP-021 rev B	10 ... 1000 mm
theta_i	Process (fluid) temperature	120	degC	spec: SPEC-M-12 §3: heating flow temperature	-50 ... 400 degC
theta_a	Ambient air temperature (also the radiation sink)	15	degC	spec: SPEC-M-12 §2: ambient design temperature	-20 ... 50 degC
lambda	Thermal conductivity of the insulation	0.04	W/(m*K)	assumption: Mineral wool; take the maker's value at the mean temperature (W3 material)	0.015 ... 0.2 W/(m*K)
h_mode	Where the outer surface coefficient comes from	convection + radiation (correlation)		assumption: Surface coefficient computed from the correlation.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
h_se	Outer surface coefficient h_se (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)	10	W/(m^2*K)	assumption: Not used: h_se is computed.	3 ... 40 W/(m^2*K)
eps	Emissivity of the outer surface (for the correlation; with h_se as given it only judges that value against condensation on a cold surface)	0.1		assumption: Bright aluminium cladding; replace with the cladding maker's emissivity.	0.05 ... 1
v_wind	Air speed over the surface (correlation; 0 = still air)	0	m/s	assumption: Still air, indoors.	0 ... 20 m/s
L_surf	Height or flow length of the flat surface (correlation, flat)	1	m	assumption: Not used: a pipe.	0.1 ... 10 m
q_max_m	Heat-loss limit per metre of pipe (generic default of 25 W/m, replace with your project limit; blank = not a criterion)	60	W/m	spec: SPEC-M-12 §5: heat loss not to exceed 60 W/m	1 ... 1000 W/m
theta_se_max	Highest surface temperature, for a hot surface (blank = not a criterion)	60	degC	spec: SPEC-M-12 §6: personnel protection, surface not above 60 °C	30 ... 70 degC
RH_a	Relative humidity of the ambient air (anti-condensation)	60	%	assumption: Not used: a hot pipe has no condensation criterion.	20 ... 95 %
t_chosen	Chosen insulation thickness (blank = report at the required thickness, no checks; with it, nothing blocks)	50	mm	spec: Pipework schedule M-SCH-021: 50 mm sections	0 ... 300 mm

3. Assumptions

A1. Steady one-dimensional heat flow through a single homogeneous layer of constant conductivity; no thermal bridges at supports, no joints, no moisture in the insulation. λ is taken at the value entered: use the value at the mean insulation temperature shown on the sheet (optionally noted in 'Temperature the conductivity applies at'). (Formula route of the standard.)
A2. With the correlation, h_se = h_c + h_r. Radiation: h_r = ε·σ·(T_s² + T_a²)(T_s + T_a) to surroundings at the ambient air temperature. Free convection to air: Holman's simplified relations at atmospheric pressure, the larger of the laminar and turbulent forms (horizontal cylinder 1.32(ΔT/D)^0.25 and 1.24ΔT^(1/3); vertical surface 1.42(ΔT/L)^0.25 and 1.31ΔT^(1/3)). Forced convection: Churchill–Bernstein for a cylinder in cross-flow, the laminar/mixed flat-plate relations along a plate; air properties from fits at the film temperature, Pr = 0.71; blended with free convection as (h_nat³ + h_forced³)^(1/3). Indicative, not the standard's own correlation. (Incropera 7th ed. §7.2, §7.4, §9.6 free convection and §9.9 mixed convection

(section numbers to be checked against the owner's copy); the free-convection constants are Holman's, Heat Transfer; the standard's own coefficients are not reproduced.)
A3. The dew point is entered (for example from B2) or is the Magnus relation at θ_a and RH, the same closed form B7 uses. Below 0 °C the relation is over ice, so RH is then taken relative to ice, whereas meteorological RH is usually relative to water: enter the dew point for sub-zero air. A margin above it is the user's. (Derived; B7 shares the function.)
A4. Pipes are treated as horizontal for free convection and flat surfaces as vertical; air properties are fits (conductivity within about 3 %, viscosity within about 4 % over 250–400 K), Pr = 0.71. (Stated.)
A5. The required thickness is the least thickness beyond which the criterion holds for every thicker layer up to 1 m, which is what matters when thin insulation on a small pipe raises the heat flow (critical diameter). (Stated.)
A6. No thickness table is used or shipped: not BS 5422, not the tables of any standard. The conductivity is the user's input, from a W3 material or the maker's data. (No standard's table is reproduced.)
A7. geometry (Surface) is taken as pipe (cylinder). (A round pipe.)
A8. lambda (Thermal conductivity of the insulation) is taken as 0.04 W/(m*K). (Mineral wool; take the maker's value at the mean temperature (W3 material))
A9. h_mode (Where the outer surface coefficient comes from) is taken as convection + radiation (correlation). (Surface coefficient computed from the correlation.)
A10. h_se (Outer surface coefficient h_se (as given; generic default of 10 W/(m²K), replace with the value for your surface; a high value is not conservative against condensation on a cold surface, use the correlation there)) is taken as 10 W/(m^2*K). (Not used: h_se is computed.)
A11. eps (Emissivity of the outer surface (for the correlation; with h_se as given it only judges that value against condensation on a cold surface)) is taken as 0.1. (Bright aluminium cladding; replace with the cladding maker's emissivity.)
A12. v_wind (Air speed over the surface (correlation; 0 = still air)) is taken as 0 m/s. (Still air, indoors.)
A13. L_surf (Height or flow length of the flat surface (correlation, flat)) is taken as 1 m. (Not used: a pipe.)
A14. RH_a (Relative humidity of the ambient air (anti-condensation)) is taken as 60 %. (Not used: a hot pipe has no condensation criterion.)

4. Working

[11] t_q = t such that |q| ≤ 60 W/m for every thicker layer — ISO 12241
= scan 0 … 1000 mm, then bisection
= 23.374 mm
Required thickness, heat-loss limit

[12] t_se = t such that □_se ≤ 60 °C for every thicker layer — ISO 12241
= scan 0 … 1000 mm, then bisection
= 7.8423 mm
Required thickness, surface-temperature limit

[13] thickness = max over the selected criteria — ISO 12241
= t_q = 23.374 mm, t_se = 7.8423 mm
= 23.374 mm
Governing required thickness (heat-loss limit)

[15] D_e = D_i + 2t
= 114 mm + 2 × 50 mm
= 214 mm
Outer diameter of the insulation, chosen thickness

[16] R_ins = ln(D_e/D_i)/(2π·□) — ISO 12241
= ln(214/114)/(2π × 0.04)
= 2.5058 m·K/W
Resistance of the insulation

[17] h_c = free convection to air, blended with forced convection — Incropera 7th ed. §7, §9.6, §9.9
= horizontal cylinder, Holman; Churchill–Bernstein if the wind blows
= 3.6748 W/(m²·K)

Convective surface coefficient at the surface temperature

[18] h_r = □·□·(T_s² + T_a²)(T_s + T_a) — Incropera 7th ed. §1.2
= □ = 0.1, □_se = 27.855 °C
= 0.58007 W/(m²·K)

Radiative surface coefficient

[19] h_se = h_c + h_r — ISO 12241
= 3.6748 + 0.58007
= 4.2549 W/(m²·K)

Outer surface coefficient used

[20] R_se = 1/(h_se·π·D_e) — ISO 12241
= 1/(4.2549 × π × 0.214 m)
= 0.34958 m·K/W

Resistance of the outer surface

[21] R = R_oth + R_ins + R_se — ISO 12241
= 0 + 2.5058 + 0.34958
= 2.8554 m·K/W

Total resistance per metre

[22] q_per_m = |□_i - □_a| / R — ISO 12241
= 105 K / 2.8554
= 36.773 W/m

Heat flow per metre of pipe (heat loss, magnitude)

[23] theta_se = □_a + q/(h_se·π·D_e) — ISO 12241
= 15 °C + 36.773 / 2.8606
= 27.855 degC

Outer surface temperature

[24] theta_m = (□_i + □_se)/2
= (120 + 27.855)/2
= 73.927 degC

Mean temperature of the insulation: take λ at this temperature

[25] D_crit = 2□/h_se at the bare surface — ISO 12241
= 2 × 0.04 / 8.1897
= 9.7683 mm

Critical diameter at the bare-surface h_se: below it, insulation raises the heat flow

[26] u_q = |q| / q_max
= |q| ≤ 60 W/m
= 0.61288

Utilisation, heat-loss limit

[27] u_se = (□_se - □_a)/(□_se,max - □_a)
= □_se ≤ 60 °C
= 0.28567

Utilisation, surface-temperature limit

5. Results

SYMBOL	VALUE	DESCRIPTION
t_q	23.374 mm	Required thickness, heat-loss limit
u_q	0.61288	Utilisation, heat-loss limit
t_se	7.8423 mm	Required thickness, surface-temperature limit
u_se	0.28567	Utilisation, surface-temperature limit
thickness	23.374 mm	Governing required thickness
q_per_m	36.773 W/m	Heat flow per metre (heat loss)
theta_se	27.855 degC	Outer surface temperature

SYMBOL	VALUE	DESCRIPTION
theta_m	73.927 degC	Mean insulation temperature
R	2.8554 m·K/W	Total resistance
D_crit	9.7683 mm	Critical diameter (at the bare-surface h_se)
h_c	3.6748 W/(m²·K)	Convective surface coefficient
h_r	0.58007 W/(m²·K)	Radiative surface coefficient
h_se	4.2549 W/(m²·K)	Outer surface coefficient
governing	"heat-loss limit"	Governing criterion

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Chosen thickness against the heat-loss limit	u_q = 0.61288	u_q ≤ 1	0.613	ISO 12241
PASS	Chosen thickness against the surface-temperature limit	u_se = 0.28567	u_se ≤ 1	0.286	ISO 12241

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: lambda (Material properties); h_se (Outer surface coefficient); eps (Surface emissivity); q_max_m, theta_se_max (Insulation criteria). Enter each as one so its source, edition and licence travel with it.

8. Verification

Template bs.insulation@1.1.0 Content hash 027d4f4ba606eddd Engine calc-engine/1.0.0
Known-answer cases passed 14/14 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0